

INFORMATION SYSTEM DEPARTMENT
INFORMATION TECHNOLOGY FACULTY – HCM
UNIVERSITY OF SCIENCE

ADVANCED DATABASE

Chapter 00

COURSE INTRODUCTION



KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN

fit@hcmus

General Information

Course:	CSC12002 – Advanced Database Systems (Cơ sở dữ liệu nâng cao)
Knowledge block:	Fundamentals of Databases
Number of credits:	4
Instructors:	<i>Theory Instructor:</i> Nguyen Tran Minh Thu, Ph.D. Vu Thi My Hang, Ph.D. <i>TA/Lab:</i> Ho Thi Hoang Vy, M.Sc. Tiet Gia Hong, M.Sc.
Contact Info:	Moodle: https://courses.fit.hcmus.edu.vn [Diễn đàn ...] Email: [ntmthu vtmhang hthvy tghong]@fit.hcmus.edu.vn <i>Please respect the following guidelines when sending your email; failure to do so may result in your email being disregarded.</i> Email Topic: [CLC-ADB-2526] ... Email Content: Must include your student code and name.

Course Objective

- Aim to equip students with advanced knowledge in the field of relational database design (collecting and analyzing user requirements, **conceptual database design**, **logical database design**, and **physical database design**).
- Focus on how to select the appropriate physical techniques (index, data fragment, partition) for efficient queries on a specific relational database management system such as MS SQL, Oracle.
- Practice reading English documents and using terminologies related to relational database design.
- Practice teamwork skills.

Teaching Plan

Tuần	CLC		Phụ trách	Đánh giá (Tại lớp)
1	Thứ hai, 06/10	Chương 1: Tổng quan về Thiết kế CSDL	VTMHang	
2	Thứ hai, 13/10	Chương 5: Chuẩn hóa	VTMHang	
3	Thứ hai, 20/10	Chương 2: Mô hình dữ liệu (ER, EER)	NTMThu	
4	Thứ hai, 27/10	Chương 3: Thiết kế quan niệm	NTMThu	
5	Thứ hai, 03/11	Chương 3: Thiết kế quan niệm (tt)	NTMThu	
6	Thứ hai, 10/11	Bài tập #1: Thiết kế quan niệm	HTH Vy	BT1
7	Thứ hai, 17/11	Chương 6: Thiết kế luận lý	VTMHang	
8	Thứ hai, 24/11	Bài tập #2: Thiết kế luận lý	HTH Vy	BT2
9	Thứ hai, 01/12	Chương 7: Thiết kế vật lý	VTMHang	
10	Thứ hai, 08/12	Bài tập #3: Thiết kế vật lý	VTMHang	BT3
11	Thứ hai, 15/12	Các chủ đề nâng cao và ôn tập	NTMThu	

Grading

Topic	Proportion (%)
Assignments (Theory Section) ▪ Quiz (Personal) ▪ Exercise/Homework (Personal/Group)	20%
(Closed Book) Exam (Theory Section)	45%
Practical Section (Project, Lab Test)	35%
Bonus	+/- 5%
Total note	100% +/- 5%

Policy and Rule

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.

References

1. *Database System: Design, Implementation, and Management*, 12th Edition, Carlos Coronel & Steven, Morris, 2015.
2. *Modern Database Management*, 13th Edition, Jeffrey A. Hoffer, V. Ramesh, Heikki Topi, 2019.
3. *Database System Concepts*, 17th Edition, Abraham Silberschatz, Henry F. Korth, S. Sudarshan, 2019
4. *Database Modeling and Design - Logical Design*, 4th Edition, Toby Teorey, Sam Lightstone, Tom Nadeau.
5. *Physical Database Design*, 4th Edition, Sam Lightstone, Toby Teorey, Tom Nadeau.

THE END

